Modeling Reasoning Dependencies for Adaptive Self-Consistency

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Abstract

Self-Consistency improves reasoning by aggregating multiple sampled responses, but incurs substantial inference cost. Adaptive self-consistency methods mitigate this cost through dynamic stopping criteria based on answer agreement or response-level confidence. However, these approaches do not explicitly account for relationships among reasoning trajectories, potentially overlooking redundant evidence arising from repeated reasoning patterns. To address this limitation, we propose a Probabilistic Graphical Model (PGM)-based framework that jointly models answer agreement, response confidence, and reasoning similarity to estimate candidate-answer correctness. By incorporating latent variables representing shared reasoning patterns, our framework performs posterior inference to assess the accumulated evidence and determine whether additional sampling is necessary. This enables both answer selection and adaptive stopping based on the inferred probability of correctness. We propose to evaluate the framework against existing adaptive self-consistency methods in terms of posterior calibration, premature stopping, and accuracy–cost trade-offs, investigating whether modeling inter-response relationships improves the reliability and efficiency of LLM reasoning.

1. Introduction

Large language models (LLMs) can solve complex reasoning problems by generating intermediate reasoning steps (Wei et al., 2022), yet a single sampled trajectory may contain errors. Self-consistency (SC) improves answer selection by sampling multiple reasoning trajectories and aggregating their final answers (Wang et al., 2023). However, its fixed sampling budget allocates the same number of generations to questions with different evidential requirements: some are resolved quickly, whereas others remain uncertain after many samples. Adaptive-Consistency (ASC) (Aggarwal et al., 2023) and Early-Stopping Self-Consistency (ESC) (Li et al., 2024) address this inefficiency by deciding when to stop from the responses observed so far.

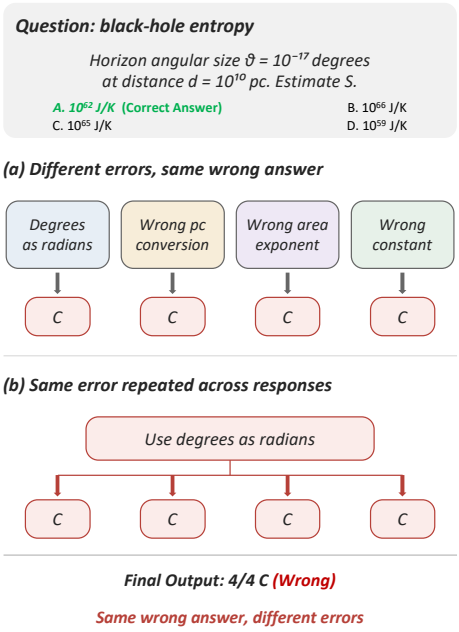


Figure 1. Illustrative failure mode of count-based adaptive sampling. Panel (a) depicts different mistakes leading to the same wrong answer; panel (b) repeats one mistake. Answer counts alone cannot distinguish these cases or certify correctness.

Existing adaptive self-consistency methods primarily rely on answer frequencies or agreement to determine when to stop sampling. However, such criteria overlook the reasoning processes underlying the sampled answers. As illustrated in Figure 1, different reasoning errors can lead to the same incorrect answer, while multiple responses may also repeat an identical mistake. Despite these distinct reasoning patterns, both cases yield identical answer counts, making them indistinguishable to count-based stopping criteria. ReASC (Kim et al., 2026) addresses this limitation in part by incorporating response-level confidence, yet it does not explicitly model relationships among reasoning trajectories. Consequently, confidence-weighted aggregation may still overlook redundant evidence arising from repeated reasoning patterns.

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This motivates considering not only the answers and confidence scores of sampled responses, but also the relationships among their reasoning trajectories. In particular, responses supporting the same answer may provide different amounts of additional evidence depending on how their reasoning overlaps. Modeling these relationships may therefore help determine both which candidate is correct and whether further sampling is necessary.

To this end, we propose a probabilistic graphical modeling framework that relates candidate-answer correctness to observed response confidence, answer agreement, and reasoning similarity, together with latent variables representing shared error or redundant support. The graph is intended as an inferential model of uncertainty about correctness, rather than an assumption that independently sampled trajectories are statistically dependent. By marginalizing over latent variables, the framework aims to estimate the posterior probability of candidate answers while accounting for overlapping evidence.

The resulting posterior informs both answer selection and adaptive stopping: the most probable candidate is selected, and sampling stops when its posterior meets a threshold calibrated on held-out validation data, subject to a maximum budget. We plan to compare against fixed-budget SC, ASC, ESC, and ReASC on reasoning tasks, assessing posterior calibration, premature stopping, and the accuracy–cost trade-off. Cost accounting will include generated tokens and the overhead of similarity computation and graphical-model inference. Ablations removing confidence, similarity, or shared-error variables will test whether the proposed structure adds value beyond confidence-aware aggregation.

Our contributions are threefold:

- We investigate inter-response relationships as additional evidence for estimating answer correctness beyond answer agreement and individual confidence.
- We propose a PGM-based framework for estimating candidate-answer posteriors and making adaptive stopping decisions.
- We establish an evaluation protocol for analyzing posterior calibration, premature stopping, and accuracy–cost trade-offs.

2. Related Works

2.1. Adaptive Self-Consistency

Self-Consistency (SC) improves reasoning reliability by sampling multiple reasoning trajectories and aggregating via majority voting, but relies on a fixed sampling budget, resulting in substantial inference cost (Wang et al., 2023). To address this limitation, adaptive self-consistency variants have adjusted the sampling budget based on observed responses (Aggarwal et al., 2023; Li et al., 2024; Wang et al., 2025). While count-based approaches rely on answer frequencies or agreement patterns, others incorporate response-level confidence into posterior estimation to guide stopping decisions (Aghazadeh et al., 2025; Kim et al., 2026). Nevertheless, these approaches do not explicitly model relationships among reasoning trajectories, potentially overlooking redundant evidence. As a result, they often lead to inefficient sampling even when sufficient evidence already exists to make a sampling decision.

2.2. PGMs for Aggregating Language Model Outputs

A probabilistic graphical model (PGM) provides a formal mathematical framework for representing conditional dependencies among random variables as a graph and factorizing the joint probability distribution according to the relationship in the graph (Jordan et al., 1999; Kschischang et al., 2001). When analyzing the inference of language models, PGMs have been adopted to aggregate multiple model outputs and select optimal answers (Arora et al., 2023; Guha et al., 2024). For instance, ConCoRD (Mitchell et al., 2022) combines confidence scores of answers generated by language models with entailment and contradiction relationships identified by a natural language inference model to improve answer accuracy and consistency through a probabilistic graphical model. While these prior efforts concentrate on aggregating or filtering predetermined predictions, our work focuses on dynamically deciding whether additional generation is required, thereby minimizing generation costs while preserving downstream task accuracy.

3. Method

3.1. Preliminaries

Problem Definition Given a question q and a language model $\mathcal{M}$, adaptive self-consistency sequentially samples responses and determines when to stop sampling and which answer to return. Let $N \geq 1$ denote the maximum number of samples. At step t, a response $r_t \sim \mathcal{M}(\cdot \mid q)$ is generated, containing a reasoning trajectory from which a final answer o_t is extracted. We denote the observed history by $H_t = (r_1, \ldots, r_t)$, an ordered sequence that contains generated responses.

A stopping rule $\delta(q, H_t) \in \{0, 1\}$ determines whether to terminate sampling based on the current history. Sampling ends at step τ for which $\delta(q, H_\tau) = 1$, or at $\tau = N$ if the stopping rule has not triggered earlier. An answer selection rule A then returns the final prediction $\hat{o} = A(q, H_\tau)$. A common choice of A is majority voting, which selects the most frequent extracted answer, with ties resolved by a fixed rule (Wang et al., 2023; Aggarwal et al., 2023). Algorithm 1

Algorithm 1 General Adaptive Self-Consistency Procedure

procedure ADAPTIVESC(q, M, N, δ, A)
  $t \leftarrow 0$, $H_0 \leftarrow ()$
  repeat
    $t \leftarrow t + 1$
    Sample a response $r_t \sim M(\cdot \mid q)$
    $H_t \leftarrow (r_1, \ldots, r_t)$
  until $\delta(q, H_t) = 1$ or $t = N$
  $\hat{o} \leftarrow A(q, H_t)$
  return $\hat{o}$
end procedure

summarizes this general procedure, leaving the stopping and answer selection rules unspecified. Our task is to design these rules to use the accumulated evidence to balance answer accuracy and sampling cost.

Probabilistic Graphical Models A Bayesian network represents a joint probability distribution over random variables $\mathbf{X} = \{X_1, \ldots, X_m\}$ using a directed acyclic graph, a directed graph with no directed cycles. Each node corresponds to a random variable, and each variable has a conditional distribution given its parents. The graph encodes conditional independence assumptions: each variable is independent of its non-descendants given its parents. For discrete random variables, these assumptions yield the factorization

$$p(\mathbf{x}) = \prod_{i=1}^{m} p\big(x_i \mid \mathbf{x}_{\mathrm{Pa}(i)}\big), \tag{1}$$

where $\mathbf{x} = (x_1, \ldots, x_m)$ is a particular assignment of values to the random variables $\mathbf{X}$, $\mathrm{Pa}(i)$ denotes the parent nodes of X_i, and $\mathbf{x}_{\mathrm{Pa}(i)}$ denotes the values assigned to the parents of X_i.

Inference uses observed variables $\mathbf{E} = \mathbf{e}$ to estimate unobserved variables $\mathbf{H}$. For a target variable $Y \in \mathbf{H}$, its posterior is

$$p(y \mid \mathbf{e}) = \frac{\sum_{\mathbf{h} \setminus y} p(y, \mathbf{h} \setminus y, \mathbf{e})}{\sum_{\mathbf{h}} p(\mathbf{h}, \mathbf{e})}, \tag{2}$$

where the numerator sums over all unobserved variables other than Y, and the denominator normalizes the distribution. Thus, a Bayesian network combines observed evidence according to its conditional independence structure to estimate uncertainty about a target variable.

3.2. PGM Graph Construction and Inference

We propose a conditional PGM that accounts for overlapping reasoning evidence when selecting an answer and deciding whether to stop sampling (Figure 2). After t generations, the observed evidence is $\mathcal{D}_t = \{(r_i, c_i, a_i, h_i)\}_{i=1}^{t}$,

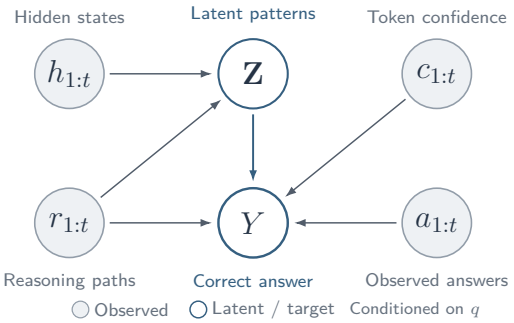


Figure 2. Conditional PGM for pattern-aware answer inference. Shaded nodes denote observed features; unshaded nodes denote latent reasoning patterns $\mathbf{Z}$ and the target answer Y. Arrows indicate conditional dependencies, with all distributions conditioned on the question q.

comprising reasoning trajectories, logit-based confidence features, canonicalized answers, and pooled hidden-state representations. A latent variable Z_i represents each response's reasoning pattern. Pattern assignments are inferred from reasoning and hidden-state features, excluding final-answer strings. Hidden states are observed inputs; reasoning pattern membership remains uncertain.

Given an assignment $\mathbf{z}$, we aggregate candidate-answer evidence within each pattern:

$$S_\theta(y; \mathcal{D}_t, \mathbf{z}) = \log p_0(y \mid q) + \sum_{k: n_k > 0} \frac{1}{n_k} \sum_{i: z_i = k} g_\theta(y, q, r_i, c_i, a_i), \tag{3}$$

where n_k is the number of responses assigned to pattern k, and g_θ is a learned support score. For fixed assignments, averaging prevents identical repetitions within one pattern from increasing evidence linearly, while distinct patterns contribute separate terms. This aggregation rule is an explicit modeling assumption to be evaluated through ablations.

The conditional model factorizes into a pattern-assignment distribution and an answer distribution. Marginalizing assignments yields

$$p_t(y) = \sum_{\mathbf{z}} p_\phi(\mathbf{z} \mid q, r_{1:t}, h_{1:t}) \times \frac{\exp S_\theta(y; \mathcal{D}_t, \mathbf{z})}{\sum_b \exp S_\theta(b; \mathcal{D}_t, \mathbf{z})}. \tag{4}$$

We train on labeled questions by minimizing the correct answer's negative log-posterior across sampled prefixes. For open-ended tasks, an OTHER state preserves probability mass for unobserved answers. Answer selection over ob-

served candidates $\mathcal{C}_t$ and stopping use the same posterior:

$$\begin{aligned} \widehat{a}_t &= \arg\max_{a \in \mathcal{C}_t} p_t(a), \\ \delta_t &= \mathbf{1}[p_t(\widehat{a}_t) \geq \tau \ \vee \ t \geq N]\,. \end{aligned} \tag{5}$$

The threshold τ is selected on held-out validation trajectories. The model represents evidential overlap without asserting dependence between independently sampled decoder outputs.

4. Experiments

4.1. Models and Baselines

Models. We evaluate two open-weight model families, each at two sizes, to assess the accuracy–cost trade-off across model families and scales.

- **Gemma 3** (Gemma Team, 2025) is a family of open-weight multimodal models developed by Google DeepMind. The models use a decoder-only Transformer with interleaved local and global attention and are trained with knowledge distillation. We use the instruction-tuned 4B and 12B variants, Gemma-3-4B-IT and Gemma-3-12B-IT.

- **Qwen3.5** (Qwen Team, 2026) is a family of open-weight vision-language models developed by the Qwen Team. Its hybrid language backbone combines Gated DeltaNet layers with gated attention. We use the 4B and 9B variants, Qwen3.5-4B and Qwen3.5-9B, with thinking mode enabled.

Baselines. We compare our framework with Adaptive-Consistency (ASC) (Aggarwal et al., 2023), Early-Stopping Self-Consistency (ESC) (Li et al., 2024), and Confidence-Guided Early Stopping (CGES) (Aghazadeh et al., 2025). ASC and ESC serve as answer-agreement baselines for evaluating the accuracy–cost trade-off. CGES provides a confidence-aware Bayesian baseline for assessing whether modeling relationships among reasoning trajectories adds value beyond confidence-based posterior aggregation. For each model, all methods will use the same prompts, generation mode, decoding settings, maximum sampling budget, and per-response token limit. We will report both the number of sampled responses and the total number of generated tokens, including Qwen's thinking tokens, as measures of generation cost.

4.2. Datasets

We evaluate mathematical reasoning, scientific question answering, and Python program synthesis on the following benchmarks.

- **MATH-500** (Hendrycks et al., 2021; Lightman et al., 2023) contains 500 competition-level mathematics problems drawn from the MATH test set, covering subjects such as algebra, geometry, and number theory. Each problem includes a reference solution and a final answer.

- **GPQA Diamond** (Rein et al., 2023) is a curated subset of GPQA containing 198 graduate-level multiple-choice questions in biology, physics, and chemistry. The questions are written and validated by domain experts, and each has four answer choices.

- **MBPP** (Mostly Basic Programming Problems) (Austin et al., 2021) contains 974 crowd-sourced Python programming tasks designed for entry-level programmers. Each task provides a natural-language description, a reference implementation, and automated tests. We use the standard test split of 500 problems.

We measure final-answer accuracy on MATH-500 and GPQA Diamond. For MBPP, we report the proportion of tasks for which the final selected program passes all provided tests.

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